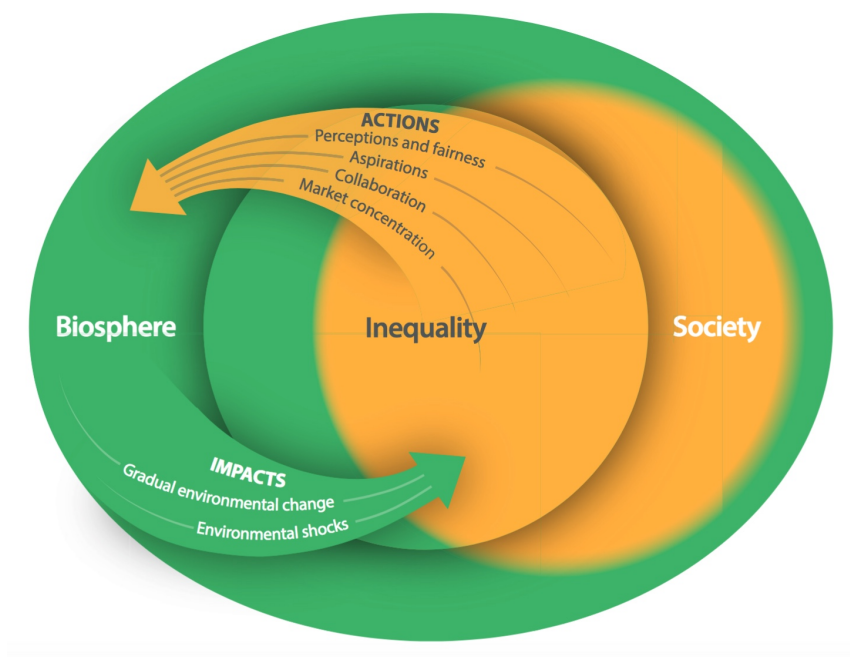


Internship report

Linking inequalities and environmental integrity at the sub-national scale of metropolitan France



[Hamann et al. \[2018\]](#)

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Abstract

Achieving sustainable development for a territory can be seen as a combination of three key elements: having sufficient wealth (1), well enough distributed for everyone to benefit from it (2), while limiting the impact on the environment (3). Thus, improving our understanding of the relationships between these parameters is crucial. This report is the first study focusing on the emerging patterns between economic inequalities and the environmental impact at a sub-national scale. To do so, we use dimensionality reduction and clustering analysis over metropolitan France's regional, departmental, and inter-communal data. In this way, we identify existing regimes that steer the relationship between an area's environmental impact and its level of inequalities. Our results provide valuable insights to address the pressing challenges of the Anthropocene.

Résumé

Le développement durable d'un territoire peut être considéré comme la combinaison de trois éléments clés : disposer de suffisamment de richesses (1), suffisamment bien réparties pour que chacun puisse en bénéficier (2), tout en limitant l'impact sur l'environnement (3). Il est donc essentiel d'améliorer notre compréhension des relations entre ces paramètres. Ce rapport est la première étude qui se concentre sur les schémas émergents entre les inégalités économiques et l'impact sur l'environnement à une échelle infranationale. Pour ce faire, nous utilisons les méthodes de réduction de dimensionnalité et d'analyse de clusters sur les données régionales, départementales et intercommunales de la France métropolitaine. Nous identifions ainsi les régimes existants qui définissent les relations entre l'impact environnemental d'une zone et son niveau d'inégalités. Nos résultats fournissent des indications précieuses pour relever les défis pressants de l'Anthropocène.

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1 Introduction

In 1972 the Club of Rome used a model to show *The Limits to Growth*. They illustrate through modeling that, in a world of finite resources, growth cannot be infinite and already explore some scenarios for the future [Meadows et al., 2013]. Decades later, environmental sciences have become a field of worldwide cooperation, leading to a remarkable research consensus over the trajectories of the Earth and its constituents. The IPCC's sixth assessment report reminds us that "Human activities, principally through emissions of greenhouse gases, have unequivocally caused global warming [and it] has led to widespread adverse impacts on food and water security, human health and on economies and society and related losses and damages to nature and people" [Lee and Romero, 2023]. This crisis is complex and already six of the nine Earth's biophysical planetary boundaries are exceeded, putting at risk its inhabitants [Richardson et al., 2023]. In an effort to safeguard a planet teeming with life, it is important to reach an equilibrium by guiding human impacts towards a sustainable path [Meadows et al., 2015]. From this necessity emerges the notion of sustainable well-being defined as:

"pursuing sustainable development to achieve well-being where it is now most conspicuously absent, as well as converting to a sustainable basis the maintenance and expansion of well-being where it already exists but is being provided by unsustainable means."
[Holdren, 2008]

This definition can be globally seen as a "triple bottom line": "to achieve high material living standards (prosperity) that are also well distributed (equality) and maintainable over the long run (environmental integrity)" [Wu et al., 2024].

Equality holds communities together and enables nations and regions to evolve along sustainable development trajectories. Inequalities, in terms of both social and natural capital, are on the rise in the world and need to be addressed as an integral part of our future on Earth [Folke et al., 2021]. Of course, it only works out with enough goods to share in-between people, and therefore sufficient prosperity. Simultaneously, environmental integrity is key for sustainability. To maintain a given prosperity the resources have to be renewed at least as fast as they are consumed in our global ecological system. There is a need to better understand the dynamics between those three parameters to reach a sustainable way for all to live on the same planet, and understanding how inequalities and the environment interact with each other is a step towards this goal. Many case studies have already shown different ways in which the environment could impact inequality levels and vice-versa (Lewis et al. [2017], Mandle et al. [2015], Husby et al. [2018]). Other ones have looked at it on a more systematic level (Hamann et al. [2018], Cushing et al. [2015], Berthe and Elie [2015]) or at a global scale (Wu et al. [2024], Grunewald et al. [2017]). Yet, none focuses on the patterns at a sub-national scale.

Our study looks at the links between inequalities and environmental integrity with a regime framework at the sub-national scale using France as an example. Our first aim is to propose a methodology derived from [Wu et al., 2024] to study regimes without a limitation on the number of variables. This will allow us to answer the following questions:

What are the main dynamics between economic inequalities and Greenhouse Gases (GHG) emissions in France between 2016 and 2021?

What are the main regimes for France at the regional, departmental, and inter-communal scale between 2016 and 2021?

Are those regimes stable across scales?

In what way certain scales can help for territory management?

This research opens the door to a series of studies on other countries to compare the patterns, starting with Chile, for which I choose here to keep the spotlight off. Combining those studies will set up a solid basis on what patterns emerge between inequalities and carbon footprint at relatively local scales, where the stakes are the highest [Bandari et al., 2024]. It will also clear up the blind spot of sub-national dynamics and highlight their similarities and differences to what we already observe at bigger scales. Decisions concerning inequalities or environmental sustainability will heavily impact the lives of people sooner or later. Understanding the contexts in which they coexist, as well as trade-offs and synergies between one another is critical for guiding public policies. And, as you will later see, some interesting patterns can emerge...

2 Methods

In order to study the interaction between economic inequalities and the environmental impact at a sub-national scale we have adapted the regime-centered methodology of a previous study looking at those interactions on an international scale [Wu et al., 2024]. Our main idea is displayed in Figure 1.

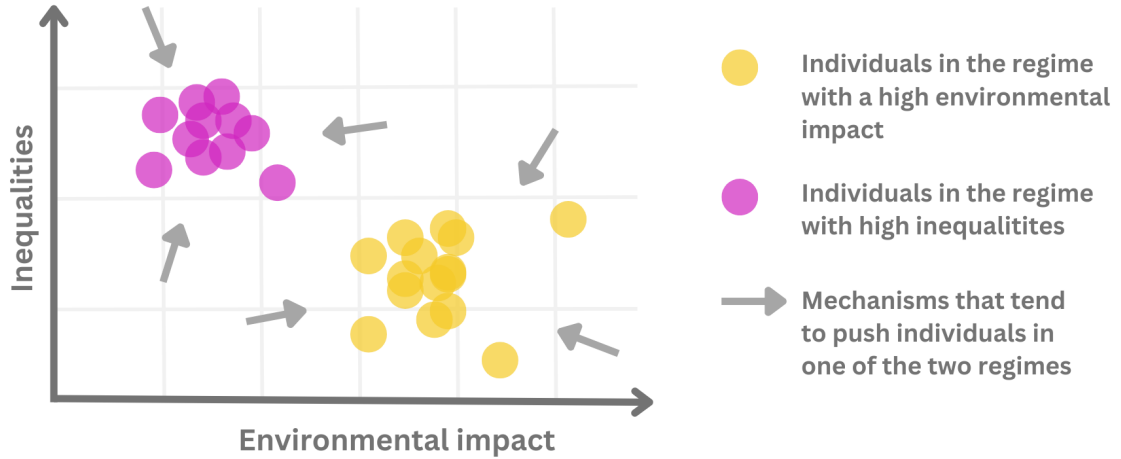


Figure 1: **Diagram of the regime framework** It is a 2D example of a regime-based analysis with two clusters. Due to underlying mechanisms, individuals are stuck into one of the two regimes, thus forming clusters.

I define an individual as a geographic area delineated by an administrative breakdown of the country, like a region, a department, or a group of municipalities for France (Appendix A.2). Depending on the scale we look at, we can have tens, hundreds, or thousands of individuals. Our main hypothesis is that mechanisms exist that tend to trap individuals within particular regimes, exposed to us in the form of clusters (Figure 1). The method aims to robustly find those regimes and define them for any number of individuals and variables. Once the regimes are established, it gives us information on the trends between our variables and what are the zones of equilibrium for our individuals.

2.1 Sub-national scale data

The first step of the methodology is to find the appropriate data covering inequalities, wealth, and environmental impact. We found those indicators on the websites of national institutions, research organisations, or through contacting researchers working on specific databases. Our

datasets consist of indicators that describe individuals over the years. An indicator is a quantitative information about the level of equality, prosperity, or environmental impact of an individual over a set of years. I call an indicator at a given year (the couple {indicator, year}) a variable. This means that if we have n_i indicators and n_t years, we end up with $N = n_i \times n_t$ variables.

Variable's Name	Description	Unit
GHG-Energy	Uncentralised energy production	$T\ CO_2\ eq/capita$
GHG-Industry	Industrial processes (except energy industry)	$T\ CO_2\ eq/capita$
GHG-Residential	Residential (with energy consumption)	$T\ CO_2\ eq/capita$
GHG-Tertiary	Service sector (with energy consumption)	$T\ CO_2\ eq/capita$
GHG-Waste	Waste	$T\ CO_2\ eq/capita$
GHG-Road	Road transport	$T\ CO_2\ eq/capita$
GHG-Other_Transport	Transport except road transport and international transport	$T\ CO_2\ eq/capita$
GHG-International_Transport	International transport	$T\ CO_2\ eq/capita$
GHG-Agriculture	Agriculture production	$T\ CO_2\ eq/capita$
GHG-Total	Total emissions (sum of all of the above)	$T\ CO_2\ eq/capita$
I-Gini	Gini index [Farris, 2010]	no unit
I-S8020	Ratio of the share of the richest 20% to that of the poorest 20%.	no unit
I-P9010	Ratio of the share of the richest 10% to that of the poorest 10%.	no unit
I-P9050	Ratio of the share of the richest 10% to that of the poorest 50%.	no unit
I-P5010	Ratio of the share of the richest 50% to that of the poorest 10%.	no unit
I-Palma	Ratio of the share of the richest 10% to that of the poorest 40%.	no unit
W-Median	Median disposable income of households	Euro (€)

Table 1: **Summary of used indicators for France multi-scale study.** *GHG* stands for per capita Greenhouse Gases emissions indicators, they are separated into 9 different sectors to better distinguish areas with the sources of their environmental impact. *I* concerns economic inequalities based on the disposable income of households. They give us different information about the distribution of wealth for households. *W* labels the wealth variable.

For the multi-scale study of France, we describe regions, departments, and Public Institution for Inter-communal Cooperation (EPCI) (Appendix A.2) with 17 indicators over 2016, 2018, and 2021. 6 of those indicators concern economic inequalities (*I*), 10 focus on Greenhouse emissions (*GHG*), and the last one is an indicator of prosperity (*W*). Economic data is from the Filosofi database [INSEE], and the territorial GHG emissions are from Citepa [GAVEL, 2019]. The time series is quite small and does not permit chronological study, though having a few years instead of only one smooths out one-year singularities and gives an idea of recent tendencies with more robustness. A complete description of the indicators we use is shown in table 1.

2.2 Trajectories study

A first step to study the regimes of individuals in a space mixing different types of indicators is to grasp an idea of the global dynamics between those types. I choose to focus on interactions between the environmental impact of an area and its economic inequalities as the influence of prosperity is not our main subject. To plot these trajectories, we select two of the indicators as representative of inequalities and environmental impact. The Gini index is a good indicator of inequalities as it accounts for economic inequalities whatever the distribution of income [Farris, 2010]. For environmental impact, Greenhouse Gases (GHG) emissions are in our case the only data available with sufficient quality. Thus, we choose the Total GHG emissions of a territory as an indicator of the environmental impact of our areas.

Projecting the trajectories of the individuals in between those two indicators gives us the global dynamic of the country at the chosen sub-scale. This information is however restricted to the general dynamics between economic inequalities and environmental impact. Additionally, we plot the densities of those trajectories using 2D kernel density estimation [Venables and Ripley, 2013] with the function "kde2d()" from the MASS R library [Ripley et al., 2020]. These graphs reveal aggregation zones where individuals tend to dwell and therefore give the main regimes in

this 2-dimensional space.

This first approach is simple and efficient as long as we focus on a low number of indicators and we have good time series. However, we need to make the most out of all the different indicators we have that may carry more detailed information, and we also do not have good and long time series (only the years 2016, 2018, and 2020). That is why in the next steps we squeeze out the patterns that can be found in all of our data using dimensionality reduction and clustering.

2.3 Dimensional reduction

The aim of the next part (Section 2.4) is to find clusters of individuals with the same behavior. However, our initial space contains $N = n_i \times n_t$ variables. It quickly happens to be a high-dimensional space (with 51 variables for France). Traditional clustering algorithms do not operate well on high dimensional space, this is known as "the curse of dimensionality" [Altman and Krzywinski, 2018]. Furthermore, some clustering algorithms have been developed to work in those conditions, but they require very-high dimensionality [Aremu et al., 2020], with often bigger spaces than ours that have an order of magnitude of 10^2 dimensions. With our middle ground, the common solution is to go from our N-dimensional space to a smaller space using a dimension reduction method, and then, apply a clustering algorithm.

We choose to use the linear method of Principal Component Analysis (PCA). It has the benefit of being a well-known and transparent method for dimension reduction, qualities that are appreciated for exploratory research. Additionally, I will go a little further by applying Multiple Factorial Analysis (MFA) due to the structure of our data instead of simple PCA. Once this step is done, we end up with a smaller space describing our individuals with fewer parameters (usually around 10 variables or so).

2.3.1 Principal Component Analysis (PCA)

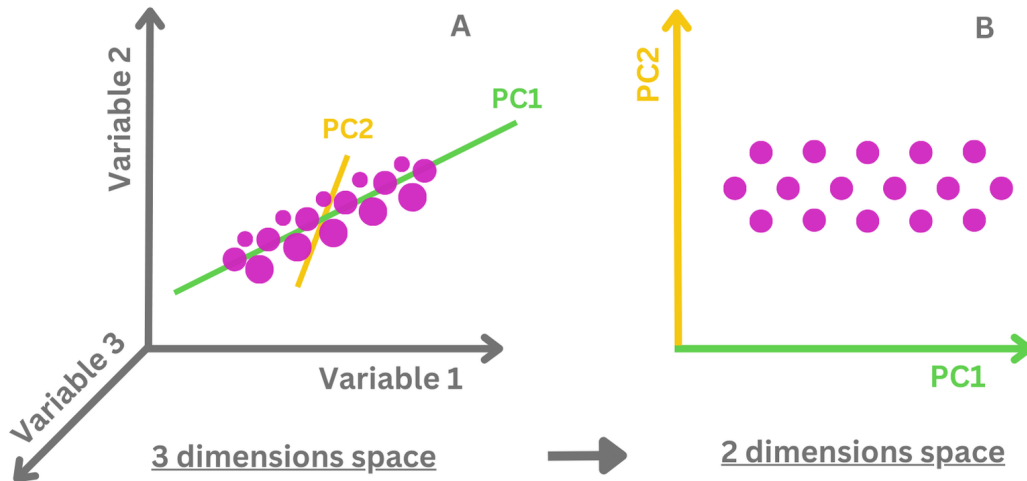


Figure 2: **Diagram of a PCA example from 3 to 2 dimensions.** On the left (A), the 3-dimensional space with the data scattered. PC1 and PC2 are the first two orthogonal directions of main deviation, also named Principal Components (PC). On the right (B), the data is represented in the 2-dimensional space made of the first two principal components. Most of the information is kept because of the initial near-flat distribution of the data.

The principle of PCA is shown in Figure 2 where PCA allows the passage from a 3-dimensional

space to a 2-dimensional one with nearly no loss of information. This is done by finding the first 2 orthogonal directions of main variation (Principal Components 1 and 2 (PC1 and PC2) on Figure 2.A), and then calculating the coordinates of the initial data in the new subspace defined by those 2 principal components (Figure 2.B).

Of course, this method isn't limited to 3-to-2-dimensional space, and can be generalized to reduce bigger spaces. As an example, in our study, we usually reduce spaces with tens of dimensions to ones between 5 and 15 dimensions. For a space with N dimensions, we can find N principal components (PCs), but every following PC contains less information than the previous one. The information is thus concentrated in the first few PCs. Retaining only those gives us a new space with fewer dimensions and nearly as much information as initially. A detailed explanation of PCA can be found in Appendix A.3.

2.3.2 Multiple Factorial Analysis (MFA)

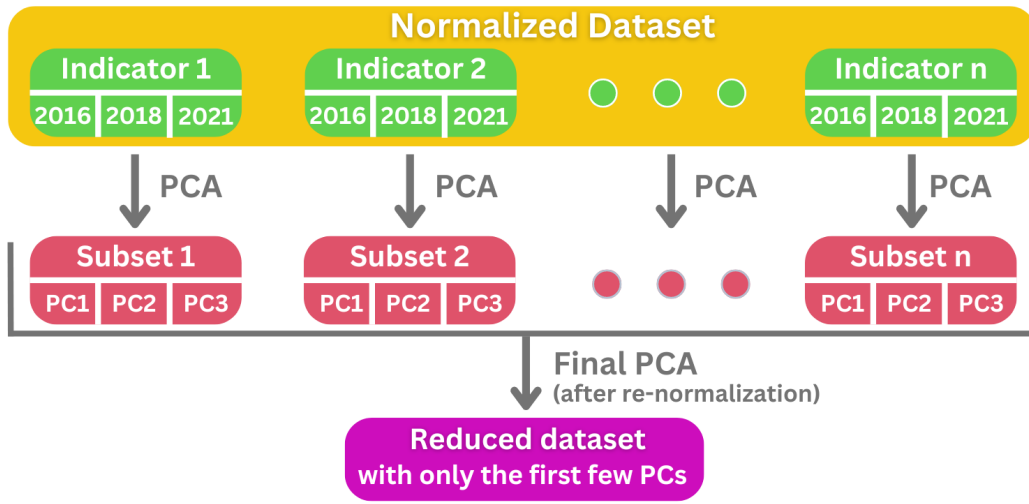


Figure 3: **Diagram of the Multiple Factorial Analysis (MFA) principle.** The initial dataset is divided into subsets of indicators. Each subset undergoes a PCA from which all PCs are preserved (no dimensional reduction). Then, one final PCA is applied to those subsets combined, from which only the first few dimensions are retained, giving us a smaller space.

The Multiple Analysis is a combination of simple PCAs as shown in Figure 3. Instead of applying a PCA over the whole dataset from the start, we apply multiple PCAs over subsets, normalize the results, and then run a final PCA over the normalized results altogether. MFA is a powerful tool that offers many advantages to bi-structured data like ours, where the subsets are naturally defined by the indicators over the different years. A detailed explanation is available in the Appendix A.4. We use the *FactoMineR* package on R to compute MFA over our data [Husson et al., 2024].

Selecting the number of Principal Components (PCs) to keep From the final PCA, we can select the first d components, reducing our space from N to d dimensions. However, choosing the number of dimensions kept is an important step as results can be rather sensitive to it. One widespread mistake in selecting the number of components to keep is the misuse of the Kaiser criterion (or Kaiser-Guttman rule) [Guttman, 1954]. It advises keeping at least as many components as there are eigenvalues above 1 for the non-reduced correlation matrix. However this criterion is widely used as a threshold (rather than a limit) to select the number of components, and I have often seen it used on the reduced correlation matrix. Moreover, it is also

a theoretical limit that has been empirically rejected due to wrong estimations and sensitivity to the number of variables, among other things [Preacher and MacCallum, 2003]. The criterion we choose to use is to retain the number of dimensions that together explain more than 95% of the variance. It ensures that most of the information is kept and always selects few enough components for our data (less than 15 usually). If too many components were to be kept with this method, other criteria exist [Preacher and MacCallum, 2003], but in my case the simpler the better.

MFA advantages over PCA Usually, the main advantage of MFA is to study the PCA results from the perspective of groups of variables. For exploratory analysis, it gives access to a whole set of information over the subsets (here the indicators) that isn't accessible with simple PCA. However, the analytical benefit of this tool is not powerful enough for our data analysis because we have too many variables. Yet, I found out that MFA consistently improved the dimension reduction efficiency by packing more information on fewer principal components. In fact, my variables are made of indicators over years, with usually highly correlated data within the variables of a given indicator. Focusing first on time is a clever way to guide the information compression where it is sure to be effective. The first PCAs of an MFA compress once each indicator data on the first PCs. Then the final PCA gets pre-ordinated data, hence leading to an improvement of the dimensional reduction.

2.4 Clustering

We now have our individuals located in a space of typically around ten or so dimensions. We call the set of individuals (which are in our case geographical areas) A . The idea of clustering is to decompose A into subsets $C_1, C_2, \dots, C_p$ such as:

$$A = \sqcup_{i=1}^p C_i \Leftrightarrow (A = \bigcup_{i=1}^p C_i \text{ and } \forall i, j \in \llbracket 1, p \rrbracket \text{ with } i \neq j : C_i \cap C_j = \emptyset)$$

To do so we assess the optimal number of clusters and then use hierarchical agglomerative clustering (HAC) improved at the end with the k-means method.

2.4.1 Finding the optimal number of clusters

An entry parameter for hierarchical agglomerative clustering (HAC) is the number of clusters. It is a key parameter as it strongly affects the results. Indeed, characterizing 10 clusters instead of 2 will not give the same outcome. In our analysis, we compare different results of different clustering over different datasets. Thus our interest in a robust and consistent way to evaluate the optimal number of clusters/regimes.

There exist many ways to assess the optimal number of clusters, but they all give very different results. We choose to estimate this number with 26 different indexes, keeping in the number of clusters considered as the best for most of the indexes [Charrad et al., 2014]. We did this using the R package NbClust [Niknafs, 2022], with the "ward.D2" method made to minimize the total within-cluster variance. Doing so gives us consistent clusters that are then characterizable, with a common ground for different datasets. It also gives us the possibility to compare my sub-national scale study to a previous international one that used the same method to determine the optimal number of regimes [Wu et al., 2024].

2.4.2 Hierarchical Agglomerative Clustering (HAC)

HAC is a pretty straightforward method. We have our set of m individuals A in our reduced space. We calculate the distance between each individual and find the individuals i_k and i_l that are the closest:

$$i_k, i_l \text{ such as } ||i_k - i_l||_2 = \min(||i_p - i_q||_2) \forall p, q \in \llbracket 1, m \rrbracket$$

Once it is done, we have our first clustering of $m - 1$ clusters, $m - 2$ being single individuals, and the last one being the couple $\{i_k, i_l\}$. We then consider our new couple as a single cluster and repeat the previous steps. Then, iteratively we get a decomposition in fewer and fewer clusters until we get only 2 of them. Then by choosing the number of clusters we want (Section 2.4.1) we extract the targeted clustering.

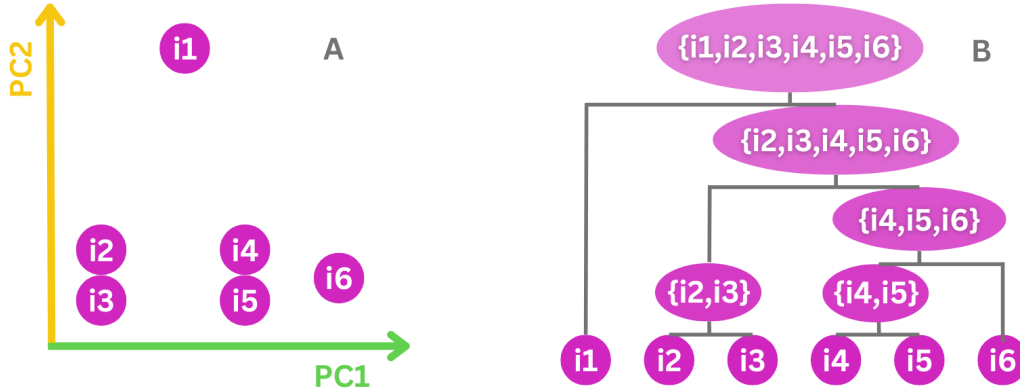


Figure 4: **Diagram of the Hierarchical Agglomerative Clustering (HAC).** On the left (A), we represent an example of the individuals' locations in a 2-dimensional space. On the right (B), the associated dendrogram that merges the cluster from bottom to top based on their distance.

A simple example is shown in Figure 4 with an initial set of $m = 6$ individuals in a 2-dimensional space. We go from 6 clusters/individuals to 4 clusters by merging i_2 with i_3 and i_4 with i_5 as they are the two couples of closest individuals. Then the closest clusters are i_6 and $\{i_4, i_5\}$, so we combine them to go from 4 to 3 clusters, and so on...

Of course, a distance between clusters needs to be defined for HAC to work. The one we use is the "ward" method which tends to minimize within-cluster variance and gives consistent clusters for interpretation. To compute the HAC clustering we use the "agnes()" function from the R package *cluster* [Maechler et al., 2023].

We compared HAC with other clustering algorithms on our data using the *clValid* R package [Brock et al., 2008]. It revealed that HAC is more stable than other algorithms on our data, hence our choice. On the other hand, HAC is a greedy algorithm with a complexity between $O(m^3)$ and $O(m^2 \log(m))$, m being the number of individuals [Manning et al., 2008]. HAC can also be rather sensitive to noise in the data. For our case, none of those are issues because m stays below 10^4 and the MFA removes most of the noise from the data.

2.4.3 Reinforcement with k-means partitioning

Once we get our clusters resulting from HAC, there is still some room for improvement using k-means clustering. This algorithm aims to minimize the within-cluster variance of an initial splitting. It is an iterative algorithm with 2 steps at each cycle. The first one is to calculate the

centroids of each cluster, the centroid e_q being the average individual of the cluster C_q :

$$e_q = \frac{1}{|C_q|} \sum_{i_j \in C_q} i_j$$

With $|C_q|$ as C_q 's cardinal number.

Then we reassign each individual to the cluster with the closest centroid :

$$C_q = \{i_j : ||i_j - e_q|| \leq ||i_j - e_k|| \ \forall k \neq q\}$$

Thereupon, iterating on it by recalculating the centroids, and redefining the clusters again and again keeps improving the partition by reducing the within-cluster variance until it stabilizes. Once the initial HAC clusters are improved with the k-means partitioning, we have our final clusters. The next and final step is to define the clusters/regimes.

2.5 Defining each cluster

Once a PCA or MFA is done, we have the coordinates of the individuals and variables in the same reduced space. This links the inputted variables between them and comparing the individuals' coordinates with the ones of the variables in the first few (commonly 2) principal components (PCs) also allows us to characterize the individuals. If we have clusters of individuals, the same reading is possible to define them by how they differentiate from the average. However, this method becomes obsolete when the number of variables becomes too important. Too many variables reduce the readability and looking at the first few PCs lose some significance. Usually, when faced with this issue, one studies the correlation between variables to remove those highly correlated and end up with fewer variables even if it can induce some information loss. Yet, the number of less correlated variables can still be too high and, above it all, our variables consist of indicators over years, applying such a method would cut through the structure of some of the indicators by removing some of the years here and there. With sorted data with a high number of variables like ours, a different methodology is necessary to define the clusters.

To define a cluster C_q we need to find which variable takes significantly different values for individuals of the cluster compared to the average individual. To better get the idea, if a cluster is only made out of one individual, we define it by the variables for which the value differs strongly for this individual from the average value. If the cluster has many individuals then it gets a bit more complex and we sort it out with the v-test defined by [Husson et al., 2017]. It corresponds to a standardized measure of the deviation of the mean of C_q individuals from the general average. The v-test can also be seen as a statistical test to check for the significance (with a p-value) of the deviation of a variable in C_q individuals in regard to all the individuals. Then, using the v-test value, we know if the tested variable has a significantly higher or lower value for C_q individuals compared to the average. A complete explanation is available in Appendix A.5. To implement the v-test I use the function "catdes()" from the R package *FactoMineR* [Husson].

3 Results

3.1 Trajectories

In order to get an initial overview of global dynamics and associated regimes between inequalities and Greenhouse Gases (GHG) emissions, we plot the individuals' trajectories and the linked densities in a 2-dimensional field defined by only two of our variables (Figure 5). The variable

kept for GHG emissions is the Total emissions because it is the one that matters in our perimeter of study to assess sustainability.

To limit global warming below 1.5°C, GHG emissions should be cut by 43% by 2030 (compared to 2019) [Lee and Romero, 2023]. Moreover, those emissions are estimated to be on a worldwide average of $4.6 \text{ T CO}_2 \text{ eq/capita}$ in 2019 [Bank, 2019]. Thus we can define $4.6 \times (1 - 0.43) = 2.6 \text{ T CO}_2 \text{ eq/capita}$ as a sustainability threshold for the next decade. Of course, this is only an order of magnitude, but it gives an idea of short-to-medium-term objectives. Comparing it to the French sub-national trajectories (Figure 5) shows how far from this goal France is, where most areas have more than 3 times those GHG emissions. The indicator selected for inequalities is the Gini indicator which has the benefit of carrying the information of economic inequalities for any distribution. Historically, a country with $\text{Gini} < 0.3$ can be seen as a country with low inequalities [Piketty and Saez, 2014]. Regarding that limit, France is a country with low inequalities and most of its areas fall below 0.3 Gini. This threshold shall also be seen as a simple order of magnitude as economic inequalities have complex fall-offs that are scale-dependent [Wilkinson, 2006].

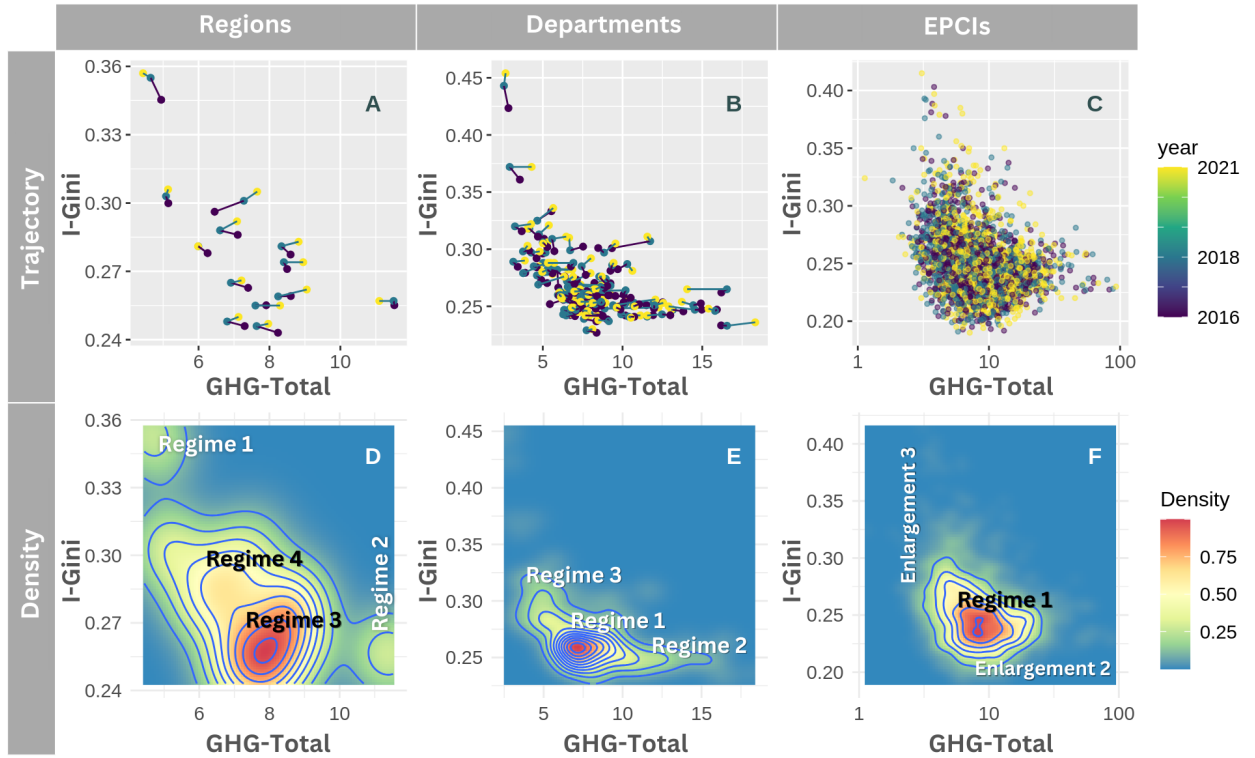


Figure 5: **Graphs of the trajectories and associated densities of different scale decomposition of France over 2016, 2018, and 2021.** The columns represent Regions (REG), Departments (DEP), and Public Institutions for Inter-communal Cooperation (EPCIs). The first row is their trajectory in the two-dimensional space of the Gini index as a function of total GHG emissions ($\text{T CO}_2 \text{ eq/capita}$). The same individual (Region or Department) points are linked by lines to better grasp the time evolution of singular areas. The second row shows the associated normalized density, revealing accretion points.

Figures 5 A, B, and C show the trajectories of respectively Regions, Departments, and Public Institutions for Inter-communal Cooperation (EPCIs). We distinguish no clear tendency over the years, which is to be expected for such a short time series (2016, 2018, and 2021). However, a relation shows up between inequalities and GHG emissions. As GHG emissions are lower, Gini index is higher, conveying higher inequalities. Inversely, as inequalities get lower, GHG

emissions are higher. This is especially true for the departmental scale (Figure 5 B), and seems to be verified for the few points of the regional scale (Figure 5 A). At the EPCI scale, the samples are more diverse, but we can still observe some boundary where a reduction in GHG emissions trades off with an increase in the lowest inequalities. Also, at this scale, an increase in GHG emissions above 10 $T\ CO_2\ eq/capita$ is also linked with an increase in the lowest existing inequalities.

3.2 Densities

The Figures 5 D, E, and F are the normalized density of Figures 5 A, B, and C. High-density zones are areas where individuals tend to dwell. They can be thought of as a set of parameters in which an individual is stable. Different accretion zones mean different stable regimes, and individuals can move between two regimes if a perturbation projects them closer to another steady zone.

At a regional scale, density interpretation is to be treated with caution as there are not many regions. Four regimes show up in this 2-dimensional space (Figure 5.D). Regime 1 is of high inequalities (Gini = 0.35) and low GHG emissions (4.5 $T\ CO_2\ eq/capita$). It is only represented by the most dense and prosperous region of France (*Île-de-France*). Regime 2 is its strict opposite with low inequalities (Gini = 0.26) and high GHG emissions (12 $T\ CO_2\ eq/capita$), and is solely made of the *Normandie* region. With low inequalities too (Gini = 0.26), but lower GHG emissions (8 $T\ CO_2\ eq/capita$), regime 3 is the biggest and concerns most of the French regions. The second biggest one is regime 4 with slightly lower emissions (7 $T\ CO_2\ eq/capita$) and slightly higher inequalities (Gini = 0.29).

At the departmental scale (Figure 5.E) fewer regimes are revealed. The main one, regime 1, is around 7 $T\ CO_2\ eq/capita$ and has low inequalities (Gini = 0.26). It corresponds to a mix between regimes 4 and 3 of the regional scale (Figure 5.D). Two other regimes are present though less defined. Regime 2 is of low inequalities (Gini = 0.25) and high GHG emissions (11 to 15 $T\ CO_2\ eq/capita$), and regime 3 is of high inequalities (Gini between 0.28 and 0.31) with low GHG emissions (5 $T\ CO_2\ eq/capita$).

At the EPCI scale, Figure 5.F has a logarithmic horizontal scale. There seems to be only one main regime (Regime 1 centered around 9 $T\ CO_2\ eq/capita$ and Gini = 0.24), but with two branches that extend top left and bottom right. These enlargements (2 and 3) seem to reflect departmental scale regimes 2 and 3 (Figure 5.E) but are even less defined. Enlargement 2 is between 15 and 80 $T\ CO_2\ eq/capita$ and around Gini = 0.25, while enlargement 3 extends from a Gini index of 0.27 to 0.35 with GHG emissions around 6 $T\ CO_2\ eq/capita$.

3.3 Clusters and regimes

We have made a first approach focusing on only two variables that represent inequalities and environmental impact. However, to better grasp the dynamics at stake it is important to make the best out of all the data we have and link them together. To do so, we apply consecutively sections 2.3, 2.4, and 2.5 methods. For the regional, departmental, and inter-communal datasets we have respectively kept the 6th, 12th, and 14th first PCs of the MFA. Specifics about the code are linked in Annex A.1.

Clusters' description Figure 6 shows the descriptions of the clusters at the three scales studied for France. When I talk about a cluster I talk about the average individual inside this cluster, but for language simplification I put an equivalence for both phrasings. In every case,

Cluster I.1				Cluster I.2				
variables	W-Median	-6.1	-5.5	-5.7	W-Median	6.1	5.5	5.7
	I-S8020	-24.7	-25.4	-24.9	I-S8020	24.7	25.4	24.9
	I-Palma	-25.2	-25.5	-25.1	I-Palma	25.2	25.5	25.1
	I-P9050	-25	-25.1	-24.7	I-P9050	25	25.1	24.7
	I-P9010	-24.3	-25.2	-24.6	I-P9010	24.3	25.2	24.6
	I-P5010	-22.4	-23.5	-23.3	I-P5010	22.4	23.5	23.3
	I-Gini	-25.9	-26.1	-25.7	I-Gini	25.9	26.1	25.7
	GHG-Total	11	10.6	11.3	GHG-Total	-11	-10.6	-11.3
	GHG-Tertiary	-7.3	-3.7	-2.2	GHG-Tertiary	7.3	3.7	2.2
	GHG-Road	10.1	10.2	10	GHG-Road	-10.1	-10.2	-10
	GHG-Residential	-11.1	-11.7		GHG-Residential	11.1	11.7	
	GHG-Other_Transp.	-3.3	-3.3	-3.7	GHG-Other_Transp.	3.3	3.3	3.7
	GHG-Int_Transp.			-2.3	GHG-Int_Transp.			2.3
	GHG-Industry	3.3	3	2.7	GHG-Industry	-3.3	-3	-2.7
	GHG-Agriculture	13.9	13.9	13.6	GHG-Agriculture	-13.9	-13.9	-13.6
	2016	2018	2021		2016	2018	2021	
	year				year			

(a) **Inter-communal clusters:** I.1 round-up 878 EPCIs while I.2 gather 335.

Cluster D.1				Cluster D.2				
variables	W-Median	-3.8	-3.8	-4	W-Median	3.8	3.8	4
	I-S8020	-6.9	-6.9	-6.9	I-S8020	6.9	6.9	6.9
	I-Palma	-6.8	-6.7	-6.7	I-Palma	6.8	6.7	6.7
	I-P9050	-6.7	-6.7	-6.5	I-P9050	6.7	6.7	6.5
	I-P9010	-6.7	-6.6	-6.7	I-P9010	6.7	6.6	6.7
	I-P5010	-6.8	-6.8	-6.9	I-P5010	6.8	6.8	6.9
	I-Gini	-7	-6.8	-6.9	I-Gini	7	6.8	6.9
	GHG-Total	5	4.6	5	GHG-Total	-5	-4.6	-5
	GHG-Tertiary	2.2	2.6	3.2	GHG-Tertiary	-2.2	-2.6	-3.2
	GHG-Road	5.2	5.3	4.6	GHG-Road	-5.2	-5.3	-4.6
	GHG-Residential	3	2.9	6.1	GHG-Residential	-3	-2.9	-6.1
	GHG-Other_Transp.	-5.6	-5.6	-5.1	GHG-Other_Transp.	5.6	5.6	5.1
	GHG-Int._Transp.	-3.5	-3.4	-3.6	GHG-Int._Transp.	3.5	3.4	3.6
	GHG-Industry	2.5			GHG-Industry	-2.5		
	GHG-Agriculture	5.2	5.2	5	GHG-Agriculture	-5.2	-5.2	-5
	2016	2018	2021		2016	2018	2021	
	year				year			

(b) **Departmental clusters:** D.1 round-up 72 departments while D.2 gather 24.

Cluster R.1				Cluster R.2				
variables	I-S8020	-2.7	-2.7	-2.7	I-S8020	2.7	2.7	2.7
	I-Palma	-2.5	-2.5	-2.5	I-Palma	2.5	2.5	2.5
	I-P9050	-2.5	-2.4	-2.4	I-P9050	2.5	2.4	2.4
	I-P9010	-2.7	-2.6	-2.6	I-P9010	2.7	2.6	2.6
	I-P5010	-2.8	-2.8	-2.8	I-P5010	2.8	2.8	2.8
	I-Gini	-2.6	-2.6	-2.6	I-Gini	2.6	2.6	2.6
	GHG-Total	2.6	2.2	2.4	GHG-Total	-2.6	-2.2	-2.4
	GHG-Tertiary	2.3	2.2		GHG-Tertiary	-2.3	-2.2	
	GHG-Road	2.1	2.1		GHG-Road	-2.1	-2.1	
	GHG-Residential	2	2.1	2.8	GHG-Residential	-2	-2.1	-2.8
	GHG-Other_Transp.	-3.1	-3.1	-2.6	GHG-Other_Transp.	3.1	3.1	2.6
	GHG-Agriculture	2.5	2.5	2.4	GHG-Agriculture	-2.5	-2.5	-2.4
	2016	2018	2021		2016	2018	2021	
	year				year			

(c) **Regional clusters:** R.1 round-up 9 regions while R.2 gather 4.

Figure 6: **Descriptive table of the optimal clusters for France at inter-communal, departmental, and regional scales.** The cells' numbers are the v-test value of each variable for the considered cluster. Only variables that significantly explain a cluster (on at least a year) are represented. Blank cells mean the values of the variable for the considered year were not significant (with a $p\text{-value} \leq 0.05$).

the optimal number of clusters is two and the clusters have strong similarities in-between the scales. For a given indicator, results are usually consistent over the 3 years of the study. Also, at a given scale, each cluster is defined with opposite characteristics to the other due to the binarity of the results.

At the Inter-communal scale (EPCIs, Figure 6a) the main cluster is I.1 with 72% of the EPCIs. It has lower prosperity with a median income lower than cluster I.2. It also has lower inequalities at all levels. And its GHG emissions are higher due to mostly higher emissions in the sectors of road transport and agriculture (and secondarily in the industry sector). On the contrary, cluster I.2 (28% of EPCIs) is more prosperous and less equal, while having lower total emissions. Yet its carbon footprint is higher in residential (for 2016 and 2018), tertiary, and off-road national transport sectors. Waste and energy sectors' carbon footprints do not characterize any of the clusters. Overall, the two clusters differ primarily in their level of economic inequalities, then by their total and sectorial carbon footprint, and eventually by their prosperity.

For the departmental scale, the division is pretty similar (Figure 6b). Cluster D.1 concerns 75% of the department. Similarly to cluster I.1, it has in order of importance, lower inequalities of every type, higher total GHG emissions, and lower prosperity. Its bigger carbon footprint is mostly due to agriculture and road transport, but is also slightly a consequence of higher emissions in the residential and tertiary sectors. Meanwhile, cluster D.2 (25% of departments) is more unequal, has less total GHG emissions, and slightly more wealth. Still, cluster D.2 has higher GHG emissions for national and international off-road transport. As for EPCIs, the energy and waste sectors' footprints are not significantly different between the 2 clusters.

At the regional scale, we again observe similar clusters with slight differences (Figure 6c). Cluster R.1 (69% of regions) is consistently more equal, with higher total GHG emissions due to (in order of importance) the agricultural, residential, tertiary, and road transport sectors. In contrast, cluster R.2 (31% of regions) has higher inequalities and lower GHG emissions. Nonetheless at this scale, the main difference between the two clusters is that cluster R.2 has a higher carbon footprint for off-road national transport. Results at the regional scale have to be put into perspective. The algorithm to find the optimal number of clusters (Section 2.4.1) gave results that were very sensitive to the scope of clusters' numbers to compare. To put it another way, the best number of clusters between 2, 3, and 4 wasn't the same as the best one between 2,3,4,5 and 6. I chose to keep 2 as the best number because it was one of the best and it was relevant seeing the departmental and inter-communal results, however, 3 and 4 clusters could also have been considered at this scale.

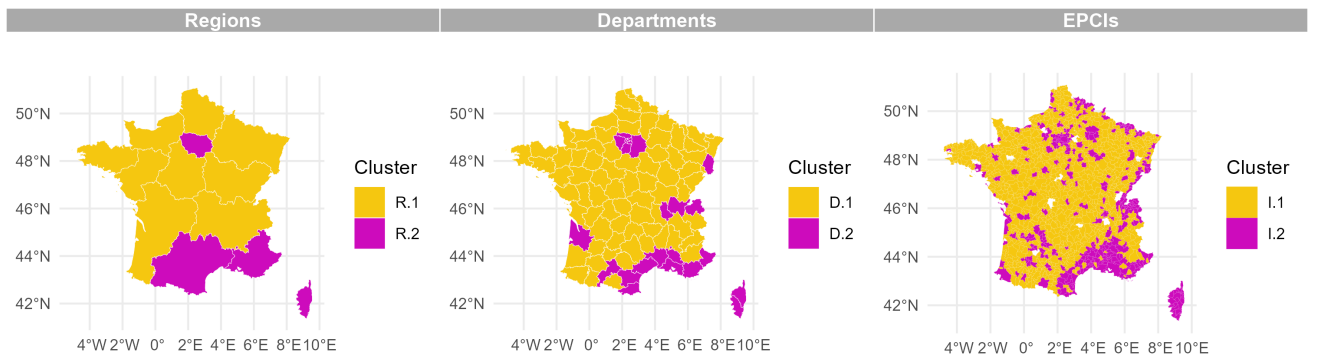


Figure 7: **Maps of the clusters for the different space scales of metropolitan France.** Similarity in colors in-between scales reflect the proximity of cluster profiles. EPCIs over which we lack data are represented in white.

Maps Figure 7 represents the breakdown of clusters over the map of metropolitan France. One piece of information that jumps to the eyes is their similarity. Wholesale, there are two types of cross-scale regimes. One is the set of R.1, D.1, and I.1 that have a bigger carbon footprint, and are less prosperous but more equal. The other is the set of R.2, D.2, and I.2 that opposes the first one with more prosperity, more inequalities, and less GHG emissions. The geographic similarities confirm previous assumptions linking those different scale clusters.

3.4 High-resolution regimes for policy advising

The following approach is an adaptation of an ecosystem-based fisheries management design [Kilborn et al., 2018]. The idea is that at the scale of action, studying the regimes gives relevant information for territory management. In our case, the targeted scale is the inter-communal one as they exist in order to manage an area. Then, instead of looking at the theoretical optimal number of clusters, we choose the number of clusters that gives us the most interesting results. It is viable because the aim is not to define the main regimes and compare them between datasets, but it is specifically aimed at territory management. In this case, more accuracy is good even if the clusters are less significant because we want to make good use of the "high" resolution. However, too many clusters and it becomes hard to interpret, too few and we lose some interesting accuracy over the regimes. Once we assess regimes, we can set up actions to go from one regime to another, or to compensate a regime backlash. Our main hypothesis here is that one area can go from one regime to the regime of another area. I choose to raise the number of clusters to six for French EPCIs. Their characterization is shown in Figure 8.

Cluster $I_{6.2}$ is the biggest cluster (39% of the EPCIs) and has the lowest level of inequalities. It also has a lower-than-average carbon footprint in nearly every sector and is more prosperous than average. $I_{6.4}$ and $I_{6.3}$ are the second and third biggest clusters with respectively 36% and 18% of the individuals. They are similar in some ways to the profiles of the 2-cluster main division found in Section 3.3. The first one ($I_{6.3}$) has much higher GHG emissions than average due to agriculture and road transport, lower prosperity, and is slightly more equal. The second one ($I_{6.4}$) has higher inequalities than average, and a lower total carbon footprint thanks to low emissions in agriculture and road transport despite having higher emissions in the residential and tertiary sectors. Its prosperity is also slightly lower than average. Cluster $I_{6.6}$ gathers 6% of the EPCIs and has similarities with $I_{6.4}$. It has a slightly lower carbon footprint than average thanks to the agricultural and road transport sector, but its main descriptor is its much higher inequalities and prosperity. One can also notice that inequalities are slightly lower for the poorest part of the distribution than for the others (P5010 v-test is around 19 while the others are consistently above 22). It means that the inequalities are even higher when we consider the top 10% or 20% of incomes. Clusters $I_{6.1}$ and $I_{6.5}$ are niches with respectively 0.5% and 0.2% of the individuals. In cluster $I_{6.5}$ the GHG emissions are higher than average as a result of much higher emissions in the non-road national and international transport. EPCIs in cluster $I_{6.1}$ have an even higher carbon footprint linked mostly to the industry and energy sectors.

Figure 9 is the map of the distribution of the 6 clusters in metropolitan France. We can link similar territories by comparing this map with Figure 7. For example, $I_{6.2}$ and $I_{6.3}$ both derive from the I.1 cluster and they concern rural territory. Yet cluster $I_{6.2}$ has interesting specifics, with lower than average GHG emissions and much lower inequalities, while $I_{6.3}$ does have slightly lower than average inequalities but a far bigger carbon footprint than average with its emissions from the road and agricultural sectors. Using this information and looking into the regimes of both clusters may give ways to push individuals from cluster $I_{6.3}$ toward the $I_{6.2}$ regime, therefore reducing their environmental impact by 6 $T CO_2 eq/capita$ (from 14 to 8 $T CO_2 eq/capita$ on average) and inequalities by 0.01 Gini (from 0.24 to 0.23 on average).

Cluster I _{6.1}				Cluster I _{6.2}				Cluster I _{6.3}												
variables	GHG-Total	20	19.9	18	variables	W-Median	4	4.7	4.7	variables	W-Median	-9.2	-9.1	-9.4						
		GHG-Int._Transp.	3.2	3.7		4.7	I-S8020	-19.7	-20.6		-20.6	I-S8020	-4.6	-4.5	-3.7					
			GHG-Industry	GHG-Residential		24.4	23.8	23.9	I-Palma		-19.8	-19.9	-19.4	I-Palma	-4.9	-4.9	-4.7			
						GHG-Other_Transp.	-19.2	-19.3	-18.8		I-P9050	-5.4	-5.3	-4.9						
						GHG-Int._Transp.	-19.5	-20	-19.8		I-P9010	-4.3	-4.6	-4						
	GHG-Energy	21.4				21	19.2	I-P5010	-19.5		-20.2	-20.5	I-P5010	-2.7	-3.2	-2.7				
	GHG-Industry	GHG-Residential		GHG-Other_Transp.		GHG-Int._Transp.	GHG-Industry	GHG-Agriculture	I-Gini		-21.6	-21.4	-20.8	I-Gini	-4.1	-4.4	-4.2			
			GHG-Waste						-2				GHG-Waste		3.2	2.8				
			GHG-Total						-5.2		-5.2	-4.2	GHG-Total	19.1	18.6	19.1				
			GHG-Tertiary						-4.7		-2.1		GHG-Tertiary	-3.9	-2.5					
GHG-Industry		GHG-Residential	GHG-Other_Transp.	GHG-Int._Transp.	GHG-Industry	GHG-Agriculture	GHG-Road	-3.9	-4		GHG-Road	18.9	19.2	14.8						
	GHG-Residential						-7.2	-7.7	-2.2	GHG-Residential	-7.3	-7.3								
	GHG-Other_Transp.						-2.4	-2.4	-2.4	GHG-Industry	3.7	3.7	3.1							
	GHG-Int._Transp.								-2.1	GHG-Agriculture	24.1	24.1	23.9							
	GHG-Industry	-2.3	-2.4	-2.5	GHG-Industry	3.7	3.7	3.1	GHG-Agriculture	24.1	24.1	23.9								
GHG-Agriculture	-3.4	-3.4	-3.3	GHG-Agriculture	24.1	24.1	23.9													
2016 2018 2021 year				2016 2018 2021 year				2016 2018 2021 year												
Cluster I _{6.4}				Cluster I _{6.5}				Cluster I _{6.6}												
variables	W-Median	-2.6	-3.1	-3.1	variables	GHG-Total	5.3	5.2	2.8	variables	W-Median	12.1	11.7	12.2						
	I-S8020	12.9	13.8	13.3			GHG-Tertiary	2				I-S8020	22.3	22	21.9					
	I-Palma	13	13	11.9				GHG-Other_Transp.	GHG-Int._Transp.		23.7	23.8	20.4	I-Palma	23	23.1	23.8			
	I-P9050	12.9	12.8	11.5							GHG-Industry	GHG-Agriculture	26.8	26.7	23.4	I-P9050	22.8	22.8	23.9	
	I-P9010	12.3	13	12.2									GHG-Industry	GHG-Agriculture	26.8	26.7	23.4	I-P9010	22.5	22.6
	I-P5010	12.8	13.8	13.8		GHG-Industry	GHG-Agriculture								26.8	26.7	23.4	I-P5010	19	19.1
	I-Gini	14.8	15	13.9					GHG-Industry						GHG-Agriculture	26.8	26.7	23.4	I-Gini	21.8
	GHG-Total	-11.1	-10.8	-11.7				GHG-Industry			GHG-Agriculture	26.8				26.7	23.4	GHG-Total	-4.9	-4.8
	GHG-Tertiary	6.1	3.1	2.6								GHG-Industry	GHG-Agriculture	26.8		26.7	23.4	GHG-Tertiary	3.3	2
	GHG-Road	-9	-9.1	-10.1		GHG-Industry	GHG-Agriculture							26.8		26.7	23.4	GHG-Road	-4.7	-4.8
GHG-Residential	10.4	10.9	2.8	GHG-Industry	GHG-Agriculture				26.8	26.7				23.4	GHG-Residential	5.7	5.8			
GHG-Industry	-3.5	-3.4	-3.1					GHG-Industry	GHG-Agriculture	26.8	26.7			23.4	GHG-Int._Transp.			2.3		
GHG-Agriculture	-12.8	-12.8	-12.9							GHG-Industry	GHG-Agriculture	26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9			GHG-Industry	GHG-Agriculture					26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9	GHG-Industry	GHG-Agriculture							26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9					GHG-Industry	GHG-Agriculture			26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9							GHG-Industry	GHG-Agriculture	26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9			GHG-Industry	GHG-Agriculture					26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9	GHG-Industry	GHG-Agriculture							26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9					GHG-Industry	GHG-Agriculture			26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9							GHG-Industry	GHG-Agriculture	26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9			GHG-Industry	GHG-Agriculture					26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9	GHG-Industry	GHG-Agriculture							26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9					GHG-Industry	GHG-Agriculture			26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9							GHG-Industry	GHG-Agriculture	26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9			GHG-Industry	GHG-Agriculture					26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9	GHG-Industry	GHG-Agriculture							26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9					GHG-Industry	GHG-Agriculture			26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9							GHG-Industry	GHG-Agriculture	26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9			GHG-Industry	GHG-Agriculture					26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9	GHG-Industry	GHG-Agriculture							26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9					GHG-Industry	GHG-Agriculture			26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9							GHG-Industry	GHG-Agriculture	26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9			GHG-Industry	GHG-Agriculture					26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9	GHG-Industry	GHG-Agriculture							26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9					GHG-Industry	GHG-Agriculture			26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9							GHG-Industry	GHG-Agriculture	26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1		
GHG-Agriculture	-12.8	-12.8	-12.9			GHG-Industry	GHG-Agriculture					26.8	26.7	23.4	GHG-Agriculture	-6.3	-6.3	-6.1</		

Figure 8: **Descriptive table of clusters for EPCIs when increased to a number of 6** Clusters $I_{6.1}$, $I_{6.2}$, $I_{6.3}$, $I_{6.4}$, $I_{6.5}$ and $I_{6.6}$ are respectively made of 6, 476, 222, 435, 3 and 71 EPCIs. The cells' numbers are the v-test value of each variable for the considered cluster. Only variables that significantly explain a cluster (on at least a year) are represented. Blank cells mean the values of the variable for the considered year were not significant (with a p -value ≤ 0.05). Variables' description is available in Table 1.

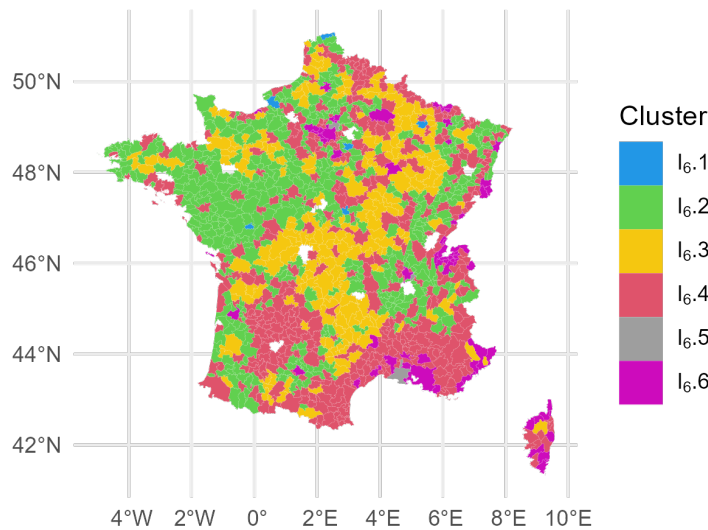


Figure 9: **Map of the inter-communal clusters for metropolitan France for 6 clusters.**

4 Discussion

4.1 Trade-off between inequalities and environmental impact

Our results show a consistent trade-off between economic inequalities and GHG emissions. Section 3.1 shows that low inequalities and low carbon footprint are difficult to achieve at the same time. If one is low enough, the other will be pretty high. The same can be seen with the regime study (Section 3.3) where the two regimes oppose inequalities and GHG emissions. They also tell us that at lower inequalities, prosperity is usually lower too, while the higher emissions are a consequence of high emissions in the agricultural and road transport sectors.

This trade-off is not limited to France. Though less pronounced, it is also visible when looking at the regional dynamic of Chile (Figure 11a in Annex A.6). Regions with lower inequalities usually have higher emissions and vice-versa. With the 1990 to 2020 time series for Chile, we also see that over time economic inequalities reduce while GHG emissions increase, though more data is needed to establish causality. At an international scale, the studies [Grunewald et al., 2017] and [Wu et al., 2024] also find a trade-off between economic inequalities and environmental impact. The last one finds this result by comparing 59 country trajectories from 1995 to 2017 (with the Gini and per-capita ecological footprint indicators). The concession between inequalities reduction and environmental footprint reduction seems to be cross-scale and independent of the studied area.

This could be the direct consequence of the fact that at two different levels of income, an increase will not have the same effect. For a high income, most of the increase in income will be saved, whereas at low income most of it will be spent, increasing economic consumption and local emissions on the way [Berthe and Elie, 2015]. Of course, it is only a first hypothesis and further study should be done to better understand this trade-off.

Synergies exist too. For instance, if we look more precisely into the EPCIs clusters (Section 3.4), the biggest regimes ($I_{6.2}$) show synergies between low inequalities and lower GHG emissions. This does not contradict previous results because the carbon footprint of EPCIs in this cluster isn't low enough to trigger the trade-off. That said, it shows that at a certain level of environmental impact, in the context of prosperous areas, very low inequalities (Gini around 0.23) are linked to reduced GHG emissions. Again, further study needs to be done to explain such a synergy but several avenues can be explored [Cushing et al., 2015].

4.2 Scale and regimes

Overall Section 3.3 shows strong scale-independent results. We find the same number of optimal regimes at the departmental and inter-communal scale, and with more uncertainty, at the regional scale too. The two regimes can be linked across scales, with I.1, D.1, and R.1 clusters that have roughly similar importance, characteristics, and location. The same can be said for I.2, D.2, and R.2. This observation is relatively good news for the sustainability study of France as it displays that the main patterns can be observed without a strong resolution, and while regional data over a set of 3 years gives results a bit too sensitive to the different parameters, the departmental data seem already sufficient to robustly describe the main dynamics.

The two regimes discovered for France reveal a strong dichotomy in the country. A comparison with a population density map of France (Appendix A.7) reveals a strong correlation between the clusters and population density. To put it another way, the first regime that concerns around 70% of metropolitan France is the rural regime. The second one with 30% of the areas can be

named the urban regime as it is mainly composed of denser territories. They reflect a strong division of the country with two different types of regimes that should be considered when aiming towards sustainable development.

The inter-communal resolution is still useful for practical applications like guiding policies (Section 3.4) but unnecessary to see the main regimes of France. Three elements shall be taken into account for a policy perspective. First, not all variables have the same room for maneuver. If there are known levers for economic inequalities [Jones and Llewellyn, 2019], it is more difficult for prosperity. For GHG emissions it depends on sectors. For instance, high emissions per capita in the agricultural sector may be a proxy of high agricultural production rather than heavily emissive agricultural methods. Actually, most of carbon footprint results need further study to take measures, while our results give clues of where to look.

Secondly, an assumption I make is that the smaller the cluster, the weaker the associated regime is likely to be. If a great bundle of EPCIs behaves alike, the dynamics gathering them may be stronger and/or wider, therefore harder to impact. To that extent, shifting EPCIs from clusters $I_{6.1}$ and $I_{6.5}$ to another may not require too much resources as they are the smallest.

Eventually, pushing an area from one regime to another existing regime is only feasible at a reasonable cost if both regimes concern territories alike. There is no point in trying to get a 200 inhabitants town towards the equilibrium of hundreds of thousands inhabitants cities.

4.3 Limits

Dimension reduction Concerning the dimensional reduction PCA (and MFA) have two major drawbacks. The first one is the need for identical time series for all individuals. This leads us to remove a big part of the data that could carry interesting information. Then, PCA is a linear method that looks at deviation to reduce the dimensions. It may leave behind nonlinear dynamics and fail to carry them in the reduced space [Hamann et al., 2018]. Other methods that are nonlinear exist to reduce a space, and some of them do not need complete time series either [Bakker, 2024]. Usually, this comes with the cost of less interpretability for the reduced space. However, the following steps of our methodology do not use the reading tools that PCA and MFA offer, which is why I would advise using different methods for dimensionality reduction once the preliminary results offered by PCA and MFA are well established.

Clusters' characterization Using the v-test to define the cluster is efficient and it is a powerful tool to do so. Yet, it is important to not consider this test as a proper statistic test (See Appendix A.5). Hypothesis aren't always verified, with too few individuals to properly use the Central Limit Theorem with non-Gaussian variables, or too many individuals to keep the p-value threshold where it stands. As a matter of fact, such a statistical test isn't great by itself to assess normality and is sensitive to the sample size [Mishra et al., 2019]. However it still gives, for a variable, the standardized deviation of the cluster mean from the global mean, with a relevant threshold to expose the variables of influence for the cluster. In this way it is powerful to sort out the variables.

Data One of the main challenges of this study was to find the appropriate data. For France and Chile territorial data is very hard to get. It is stored in the native language of the country, and dispersed over many different datasets. The data are never consistent over the years as methods change and the country's breakdown too. For instance, both France and Chile have had regional changes over the last decades that affect the way data is collected, but the harmonization is not done beforehand and is sometimes impossible. France doesn't have good historical inequality data but [Bonnet and d'Albis, 2022] should publish a new database fixing

this issue in 2025. Still, territorial data for Greenhouse Gases emissions aren't estimated before 2016. Other environmental measures are not exploitable either due to the data quality, or due to their lack of variation over the small period of the study.

We lead our approach to sustainability with a territorial perspective. However, it has a strong limit in our interdependent societies. Exported and imported GHG emissions aren't taken into account which also means that high-income people's impact on the environment is usually underestimated due to investments abroad. As reduced inequalities may be negatively linked to increased carbon footprint on the territory, it may not be the same with a global balance sheet. Of course, despite this emissions leakage, territorial study still gives a lot of relevant information about the dynamics of a territory.

A lot of work needs to be done regarding the monitoring of countries' dynamics and their impact on the environment. As it is, indicators are poorly chosen, not standardized across countries, and the progress toward Sustainable Development Goals (SDGs) is seldom measured. It is true at a national scale but even more at lower resolutions where most of the data isn't available and territories that need it the most do not have any. For example, France has a huge data gap for its overseas departments, regions, and communities (DROM-COM) concerning inequalities and the environmental impact. Our study focuses on SDGs 5 (Gender inequality), 10 (Economic inequality), 14 (Life below water) and 15 (Life on land). The assessment I have made concerning data availability at the sub-national scale for France and Chile will be used by Meike Hamann and the Beijer Young Scholars [Folke et al., 2020] to audit the state of data measurement and guide the planned 2030 SDG review.

5 Conclusion

This study highlights the critical interplay between economic inequalities and environmental impacts at a sub-national scale, with the example of metropolitan France. We find a global trade-off between inequalities and GHG emissions. Furthermore, the two main regimes consistently confront one another across the sub-national scales of France, opposing in the same way equality and sustainability. The difficulty to reach at the same time low inequalities and a low carbon footprint needs to be taken into account when aiming to influence one or the other. Looking deeper into the regimes we also show that in the right conditions, a synergy exists between reducing inequalities and GHG emissions. Moving forward, it is essential to integrate these findings into policy frameworks to promote sustainable development, ensuring that both prosperity and environmental integrity are achieved for all communities. Looking out for integrating other types of inequalities (like gender or political-based inequalities), along with other environmental indicators (focusing on biodiversity or water for instance) would further improve the results and their consequences. We hope to realize the same type of sub-national analysis on a whole set of countries to see the discrepancies depending on the country's profile.

References

- Hervé Abdi and Dominique Valentin. Multiple Factor Analysis (MFA). *Encyclopedia of Measurement and Statistics*, 2007. URL <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=1d3c368f833a726a3f294ad90f7c03772f7e3a36>.
- Hervé Abdi and Lynne J. Williams. Principal component analysis. *2010 John Wiley & Sons, Inc.*, 2:433–459, August 2010. URL <https://personal.utdallas.edu/~herve/abdi-awPCA2010.pdf>.
- Naomi Altman and Martin Krzywinski. The curse(s) of dimensionality. *Nature Methods*, 15(6):399–400, June 2018. ISSN 1548-7105. doi: 10.1038/s41592-018-0019-x. URL <https://www.nature.com/articles/s41592-018-0019-x>. Publisher: Nature Publishing Group.
- Oluseun Omotola Aremu, David Hyland-Wood, and Peter Ross McAree. A machine learning approach to circumventing the curse of dimensionality in discontinuous time series machine data. *Reliability Engineering & System Safety*, 195:106706, March 2020. ISSN 0951-8320. doi: 10.1016/j.res.2019.106706. URL <https://www.sciencedirect.com/science/article/pii/S0951832019304752>.
- Jonathan D. Bakker. Comparison of Ordination Techniques. January 2024. URL <https://uw.pressbooks.pub/appliedmultivariatestatistics/chapter/comparison-of-ordination-techniques/>. Book Title: Applied Multivariate Statistics in R Publisher: University of Washington.
- Reihaneh Bandari, Enayat A. Moallemi, Ali Kharrazi, Robert Šakić Trogrlić, and Brett A. Bryan. Transdisciplinary approaches to local sustainability: aligning local governance and navigating spillovers with global action towards the Sustainable Development Goals. *Sustainability Science*, 19(4):1293–1312, July 2024. ISSN 1862-4057. doi: 10.1007/s11625-024-01494-6. URL <https://doi.org/10.1007/s11625-024-01494-6>.
- World Bank. World Bank Open Data, 2019. URL <https://data.worldbank.org>.
- Alexandre Berthe and Luc Elie. Mechanisms explaining the impact of economic inequality on environmental deterioration. *Ecological Economics*, 116:191–200, August 2015. ISSN 0921-8009. doi: 10.1016/j.ecolecon.2015.04.026. URL <https://www.sciencedirect.com/science/article/pii/S0921800915002116>.
- Florian Bonnet and Hippolyte d’Albis. The French Regional Database - French Regional Database, 2022. URL <https://frdata.org/en/>.
- Guy Brock, Vasyli Pihur, Susmita Datta, and Somnath Datta. clValid: An R Package for Cluster Validation. March 2008. URL <https://www.jstatsoft.org/article/view/v025i04>.
- Malika Charrad, Nadia Ghazzali, Véronique Boiteau, and Azam Niknafs. NbClust: An R Package for Determining the Relevant Number of Clusters in a Data Set. *Journal of Statistical Software*, 61:1–36, November 2014. ISSN 1548-7660. doi: 10.18637/jss.v061.i06. URL <https://doi.org/10.18637/jss.v061.i06>.
- Lara Cushing, Rachel Morello-Frosch, Madeline Wander, and Manuel Pastor. The Haves, the Have-Nots, and the Health of Everyone: The Relationship Between Social Inequality and Environmental Quality. *Annual Review of Public Health*, 36 (Volume 36, 2015):193–209, March 2015. ISSN 0163-7525, 1545-2093. doi: 10.1146/annurev-publhealth-031914-122646. URL <https://www.annualreviews.org/content/journals/10.1146/annurev-publhealth-031914-122646>. Publisher: Annual Reviews.

- Frank A. Farris. The Gini Index and Measures of Inequality. *The American Mathematical Monthly*, 117(10):851–864, December 2010. ISSN 0002-9890. doi: 10.4169/000298910X523344. URL <https://www.tandfonline.com/doi/abs/10.4169/000298910X523344>. Publisher: Taylor & Francis.
- Carl Folke, Anne-Sophie Crépin, Emilie Lindkvist, Juan Rocha, Caroline Schill, and Patrik J.G. Henriksson. Inequality and the Biosphere Project, October 2020. URL <https://bys2group.wordpress.com/sdg-grant/>.
- Carl Folke, Stephen Polasky, Johan Rockström, Victor Galaz, Frances Westley, Michèle Lamont, Marten Scheffer, Henrik Österblom, Stephen R. Carpenter, F. Stuart Chapin, Karen C. Seto, Elke U. Weber, Beatrice I. Crona, Gretchen C. Daily, Partha Dasgupta, Owen Gaffney, Line J. Gordon, Holger Hoff, Simon A. Levin, Jane Lubchenco, Will Steffen, and Brian H. Walker. Our future in the Anthropocene biosphere. *Ambio*, 50(4):834–869, April 2021. ISSN 1654-7209. doi: 10.1007/s13280-021-01544-8. URL <https://doi.org/10.1007/s13280-021-01544-8>.
- Antoine GAVEL. Inventaires spatialisés, September 2019. URL <https://www.citepa.org/fr/inventaire-spatialise/>.
- Nicole Grunewald, Stephan Klasen, Inmaculada Martínez-Zarzoso, and Chris Muris. The Trade-off Between Income Inequality and Carbon Dioxide Emissions. *Ecological Economics*, 142: 249–256, December 2017. ISSN 0921-8009. doi: 10.1016/j.ecolecon.2017.06.034. URL <https://www.sciencedirect.com/science/article/pii/S092180091630619X>.
- Louis Guttman. Some necessary conditions for common-factor analysis. *Psychometrika*, 19(2): 149–161, June 1954. ISSN 1860-0980. doi: 10.1007/BF02289162. URL <https://doi.org/10.1007/BF02289162>.
- Maike Hamann, Kevin Berry, Tomas Chaigneau, Tracie Curry, Robert Heilmayr, Patrik J.G. Henriksson, Jonas Hentati-Sundberg, Amir Jina, Emilie Lindkvist, Yolanda Lopez-Maldonado, Emmi Nieminen, Matías Piaggio, Jiangxiao Qiu, Juan C. Rocha, Caroline Schill, Alon Shepon, Andrew R. Tilman, Inge van den Bijgaart, and Tong Wu. Inequality and the Biosphere. *Annual Review of Environment and Resources*, 43(1): 61–83, 2018. doi: 10.1146/annurev-environ-102017-025949. URL <https://doi.org/10.1146/annurev-environ-102017-025949>. _eprint: <https://doi.org/10.1146/annurev-environ-102017-025949>.
- John P. Holdren. Science and Technology for Sustainable Well-Being. *Science*, 319(5862):424–434, January 2008. doi: 10.1126/science.1153386. URL <https://www.science.org/doi/10.1126/science.1153386>. Publisher: American Association for the Advancement of Science.
- Trond Husby, Henri L. F. de Groot, Marjan W. Hofkes, and Tatiana Filatova. Flood protection and endogenous sorting of households: the role of credit constraints. *Mitigation and Adaptation Strategies for Global Change*, 23(2):147–168, February 2018. ISSN 1573-1596. doi: 10.1007/s11027-015-9667-7. URL <https://doi.org/10.1007/s11027-015-9667-7>.
- Francois Husson. catdes: Categories description in FactoMineR: Multivariate Exploratory Data Analysis and Data Mining. URL <https://rdrr.io/cran/FactoMineR/man/catdes.html>.
- Francois Husson, Sebastien Le, and Jérôme Pagès. *Exploratory Multivariate Analysis by Example Using R*. Chapman and Hall/CRC, New York, 2 edition, May 2017. ISBN 978-0-429-22543-7. doi: 10.1201/b21874.

- Francois Husson, Julie Josse, Sebastien Le, and Jeremy Mazet. FactoMineR: Multivariate Exploratory Data Analysis and Data Mining, April 2024. URL <https://cran.r-project.org/web/packages/FactoMineR/index.html>.
- INSEE. Revenus localisés sociaux et fiscaux | Insee. URL <https://www.insee.fr/fr/metadonnees/source/serie/s1172>.
- Russell Jones and John Llewellyn. Reducing Inequalities. *National Institute Economic Review*, 250(1):R75–R82, November 2019. ISSN 0027-9501. doi: 10.1177/002795011925000121. URL <https://doi.org/10.1177/002795011925000121>. Publisher: SAGE Publications Ltd.
- Joshua P. Kilborn, Michael Drexler, and David L. Jones. Fluctuating fishing intensities and climate dynamics reorganize the Gulf of Mexico’s fisheries resources. *Ecosphere*, 9(11):e02487, 2018. ISSN 2150-8925. doi: 10.1002/ecs2.2487. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/ecs2.2487>. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/ecs2.2487>.
- Hoesung Lee and José Romero. IPCC, 2023: Climate Change 2023: Synthesis Report, Summary for Policymakers. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds.)]. IPCC, Geneva, Switzerland. 2023. URL <https://doi.org/10.59327/IPCC/AR6-9789291691647.001>. Num Pages: 42 Pages: 1-34 Place: Geneva, Switzerland Publisher: Intergovernmental Panel on Climate Change (IPCC).
- Joshua A. Lewis, Wayne C. Zipperer, Henrik Ernstson, Brittany Bernik, Rebecca Hazen, Thomas Elmqvist, and Michael J. Blum. Socioecological disparities in New Orleans following Hurricane Katrina. *Ecosphere*, 8(9):e01922, 2017. ISSN 2150-8925. doi: 10.1002/ecs2.1922. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/ecs2.1922>. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/ecs2.1922>.
- Martin Maechler, Peter Rousseeuw (Fortran original), Anja Struyf (S original), Mia Hubert (S original), Kurt Hornik [trl, ctb] (port to R; maintenance(1999 2000)), Matthias Studer, Pierre Roudier, Juan Gonzalez, Kamil Kozłowski, Erich Schubert (fastpam options for pam()), and Keefe Murphy (volume.ellipsoid({d >= 3})). cluster: "Finding Groups in Data": Cluster Analysis Extended Rousseeuw et al., December 2023. URL <https://cran.r-project.org/web/packages/cluster/index.html>.
- Lisa Mandle, Heather Tallis, Leonardo Sotomayor, and Adrian L Vogl. Who loses? Tracking ecosystem service redistribution from road development and mitigation in the Peruvian Amazon. *Frontiers in Ecology and the Environment*, 13(6):309–315, 2015. ISSN 1540-9309. doi: 10.1890/140337. URL <https://onlinelibrary.wiley.com/doi/abs/10.1890/140337>. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1890/140337>.
- Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. Hierarchical clustering. In *Introduction to Information Retrieval*. Cambridge University Press, 2008. URL <https://nlp.stanford.edu/IR-book/html/htmledition/hierarchical-clustering-1.html>.
- Donella H. Meadows, Jorgen Randers, and Dennis L. Meadows. The Limits to Growth (1972). In *The Limits to Growth (1972)*, pages 101–116. Yale University Press, October 2013. ISBN 978-0-300-18847-9. doi: 10.12987/9780300188479-012. URL <https://www.degruyter.com/document/doi/10.12987/9780300188479-012/html>.
- Donella H. Meadows, Dennis L. Meadows, Jørgen Randers, and William W. Behrens. The Limits To Growth. In *Green Planet Blues*. Routledge, 5 edition, 2015. ISBN 978-0-429-49374-4. Num Pages: 5.

- Prabhaker Mishra, Chandra M Pandey, Uttam Singh, Anshul Gupta, Chinmoy Sahu, and Amit Keshri. Descriptive Statistics and Normality Tests for Statistical Data. *Annals of Cardiac Anaesthesia*, 22(1):67–72, 2019. ISSN 0971-9784. doi: 10.4103/aca.ACA_157_18. URL <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6350423/>.
- Malika Charrad and Nadia Ghazzali and Veronique Boiteau and Azam Niknafs. NbClust: Determining the Best Number of Clusters in a Data Set, May 2022. URL <https://cran.r-project.org/web/packages/NbClust/index.html>.
- Jérôme Pagès. *Multiple Factor Analysis by Example Using R*. CRC Press, 2014. ISBN 978-1-4822-0547-3. URL https://books.google.co.uk/books?hl=fr&lr=&id=BR5jDAAAQBAJ&oi=fnd&pg=PP1&dq=multiple+factor+analysis+by+example&ots=isMsOLcAB7&sig=LNKo9I8QCT2KizJJQtQBYyU2k1Y&redir_esc=y#v=onepage&q=multiple%20factor%20analysis%20by%20example&f=false.
- Thomas Piketty and Emmanuel Saez. Inequality in the long run. *Science*, 344(6186):838–843, May 2014. doi: 10.1126/science.1251936. URL <https://www.science.org/doi/10.1126/science.1251936>. Publisher: American Association for the Advancement of Science.
- Kristopher J. Preacher and Robert C. MacCallum. Repairing Tom Swift’s Electric Factor Analysis Machine. *Understanding Statistics*, 2(1):13–43, 2003. URL https://www.quantpsy.org/pubs/preacher_maccallum_2003.pdf.
- Katherine Richardson, Will Steffen, Wolfgang Lucht, Jørgen Bendtsen, Sarah E. Cornell, Jonathan F. Donges, Markus Drüke, Ingo Fetzer, Govindasamy Bala, Werner von Bloh, Georg Feulner, Stephanie Fiedler, Dieter Gerten, Tom Gleeson, Matthias Hofmann, Willem Huiskamp, Matti Kummu, Chinchu Mohan, David Nogués-Bravo, Stefan Petri, Miina Porkka, Stefan Rahmstorf, Sibyll Schaphoff, Kirsten Thonicke, Arne Tobian, Vili Virkki, Lan Wang-Erlandsson, Lisa Weber, and Johan Rockström. Earth beyond six of nine planetary boundaries. *Science Advances*, 9(37):eadh2458, September 2023. doi: 10.1126/sciadv.adh2458. URL <https://www.science.org/doi/full/10.1126/sciadv.adh2458>. Publisher: American Association for the Advancement of Science.
- Brian Ripley, Bill Venables, Douglas M. Bates, Kurt Hornik, Albrecht Gebhardt, and David Firth. Support Functions and Datasets for Venables and Ripley’s MASS. *R Package*, April 2020. URL <http://download.nust.na/pub3/cran/web/packages/MASS/MASS.pdf>.
- Jonathon Shlens. A tutorial on Principal Component Analysis. *Google Research*, April 2014. URL <https://arxiv.org/pdf/1404.1100>.
- Martin H. Trauth. Multivariate Statistics. In Martin H. Trauth, editor, *Python Recipes for Earth Sciences*, pages 381–428. Springer International Publishing, Cham, 2022. ISBN 978-3-031-07719-7. doi: 10.1007/978-3-031-07719-7_9. URL https://doi.org/10.1007/978-3-031-07719-7_9.
- W. N. Venables and B. D. Ripley. *Modern Applied Statistics with S-PLUS*. Springer Science & Business Media, April 2013. ISBN 978-1-4757-3121-7. Google-Books-ID: tovgBwAAQBAJ.
- Richard G. Wilkinson. The Impact of Inequality. *Social Research: An International Quarterly*, 73(2):711–732, 2006. ISSN 1944-768X. URL <https://muse.jhu.edu/pub/1/article/527475>. Publisher: Johns Hopkins University Press.
- Tong Wu, Juan C. Rocha, Kevin Berry, Tomas Chaigneau, Maike Hamann, Emilie Lindkvist, Jiangxiao Qiu, Caroline Schill, Alon Shepon, Anne-Sophie Crépin, and Carl Folke. Triple

Bottom Line or Trilemma? Global Tradeoffs Between Prosperity, Inequality, and the Environment. *World Development*, 178:106595, June 2024. ISSN 0305-750X. doi: 10.1016/j.worlddev.2024.106595. URL <https://www.sciencedirect.com/science/article/pii/S0305750X24000652>.

A Appendix

A.1 Code and Data

The datasets and the code used to produce the datasets and the results are available on <https://github.com/MathisMallet/Inequality-and-Biosphere>

It also contains all the datasets, code, and results for Chile.

To get the original data you can contact me at: mathis.mallet@ens.psl.eu

A.2 France's breakdowns

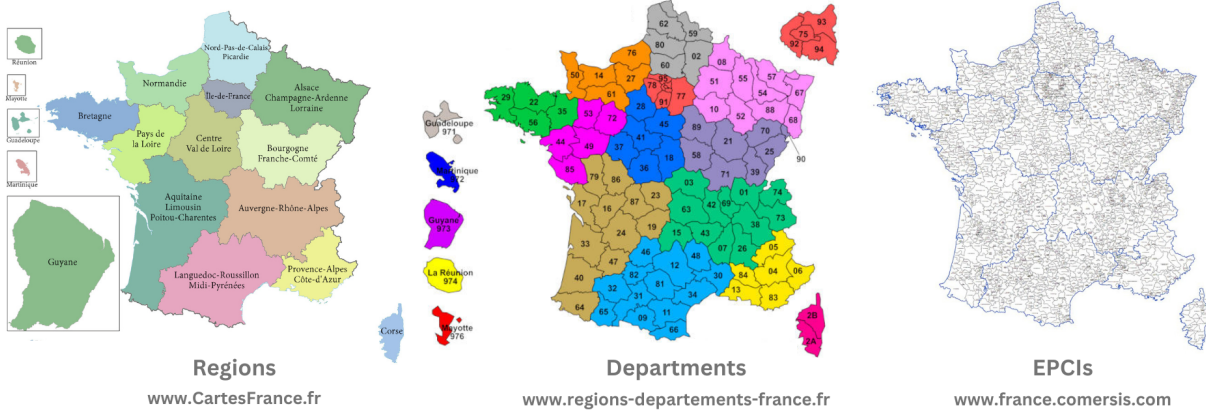


Figure 10: **Maps of the different studied breakdowns of France** Out of the 18 French regions, we have data only for the 13 metropolitan. Likewise, out of the 101 French departments, we only study the 90 metropolitan departments. Lastly, we managed to get data for 1213 of the 1253 EPCIs.

A.3 PCA calculation

Standardization The first step before doing a PCA is to standardize the data. This is common practice in machine learning as you need comparable data in order to extract information from it. Because PCA looks at the directions of main variation, we first need to center every variable to look at the variations from the mean and not at the values themselves. Then, due to the different nature of our variables, we transform the data at a comparable scale by dividing the variations of each variable by the standard deviation (σ) of its values. This step is important to avoid domination from data with higher ranges with initially bigger standard deviations over the ones with smaller ranges. Let v be the initial data for one variable, a vector with the K values of the variable (one for each individual). We define $\bar{v}$ as its mean and σ as its standard deviation. Then the standardized data $\hat{v}$ is :

$$\hat{v} = \frac{v - \bar{v}}{\sigma}$$

With $E(\hat{v}) = 0$ and $\sigma_{\hat{v}} = 1$

Once we apply this standardization to each variable's values we can pass to the next step and compare their deviations.

Finding the principal components We structure the data as a matrix X with N columns of (standardized) variables and K rows of individuals described by those variables:

$$X_{n \times N} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1N} \\ x_{21} & x_{22} & \cdots & x_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mN} \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}^T$$

Finding the first PC is finding the direction of main variation. It is equivalent to finding a unit vector a_1 so that the projection of X over a_1 has a maximal variance. This projection is $\Pi_{a_1}(X) = X \cdot a_1$. Because the initial data is centered and reduced, the variance of the projection is:

$$\frac{1}{K} \Pi_{a_1}(X)^T \Pi_{a_1}(X) = \frac{1}{K} a_1^T X^T X a_1 = a_1^T C a_1$$

With C the covariance matrix of the initial data $C = \frac{1}{K} X^T X$.

Maximizing the variance $\Pi_{a_1}(X) = X \cdot a_1$ with $\|a_1\| = 1$ is equivalent to a constrained optimization problem that can be solved with a Lagrangian multiplier. We denote $L(a_1, \lambda_1)$ the Lagrangian such as:

$$L(a_1, \lambda_1) = a_1^T C a_1 - \lambda_1 (a_1^T a_1 - 1)$$

And $a_1^T C a_1$ reach a local extreme if and only if $\frac{\partial L}{\partial a_1} = 0$ Moreover $\frac{\partial L}{\partial \lambda_1} = (C + C^T)a_1 - 2\lambda_1 a_1$

And because C is symmetrical: $\frac{\partial L}{\partial a_1} = 2C a_1 - 2\lambda_1 a_1$

Meaning $\frac{\partial L}{\partial a_1} = 0 \Leftrightarrow C a_1 = \lambda_1 a_1$ (1)

Which is equivalent to a_1 being an eigenvector of C with the associated eigenvalue λ_1 .

Taking into account that the second Lagrangian condition is $\frac{\partial L}{\partial \lambda_1} = 0 \Leftrightarrow \|a_1\| = 1$ we get from equation (1): $a_1^T C a_1 = \lambda_1 \|a_1\| = \lambda_1$ meaning that the highest value for $a_1^T C a_1$ is the highest eigenvalue of C . Similarly, we can deduce the next PCs, with the second maximum being the second highest eigenvalue, the third one the third eigenvalue, etc. And the associated eigenvectors being the principal components.

Finding those components of maximum variance can therefore be done by diagonalizing C to find its associated eigenvalues (as a covariance matrix C is a positive real matrix, meaning it is diagonalizable thanks to the spectral theorem). However, a more efficient approach is to deduce the PCs from a Singular Value Decomposition, equivalence is demonstrated in Annex (refannex:à faire).

Once the PCA is done we extract the N Principal Components (PCs) $\{a_1, a_2, \dots, a_N\}$ as a new complete base, and the coordinates of our individuals in this new base. More detailed calculation is explained by [Shlens, 2014], a complete explanation can be found in [Abdi and Williams, 2010], and [Trauth, 2022] show examples in Earth Sciences using python.

A.4 MFA calculation

To do the first PCAs we normalize the data as explained in Section A.3. Then we separate our data into the natural n_i subsets that are the indicators. Each subset is, therefore, the data for one indicator, and has n_t variables (the values of the indicators over the n_t years. Then we run a PCA over each subset and recover all the PCs. In this step, there is no reduction of the dimension, but for each subset the data is now in a new ordination base that reveals the directions of main variation for the subsets.

To normalize each subset's PCA result we divide the new coordinates matrix by its first singular values. The first singular value is the square root of the first eigenvalue (the one with

the highest value). This step makes PCA results comparable to each other.

Then we apply a final PCA over the merged dataset of the normalized subsets' PCA results. A more detailed explanation is given by [Abdi and Valentin, 2007] and an example in R is available in [Pagès, 2014].

A.5 v-test

We consider a cluster $C_q \subset A$ with $|C_q|$ individuals. The total number of individuals is $|A|$. We consider a variable V with an average over A of $\bar{v}$ and a variance of σ^2 . The variable $\bar{v}_q$ is the average of V for the individuals in C_q .

Our null hypothesis in this test is that the values of V for individuals of cluster C_q are randomly drawn from all the possible values of V . Considering $\bar{v}_q$ as the random variable average of the C_q individual and with the previous hypothesis:

$$E(\bar{V}_q) = \bar{v} \text{ and } V(\bar{V}_q) = \frac{\sigma^2(|A| - |C_q|)}{|C_q|(|A| - 1)}.$$

The v-test formula is defined as below:

$$v - test = \frac{\bar{v}_q - \bar{v}}{\sqrt{\frac{\sigma^2}{|C_q|} \left(\frac{|A| - |C_q|}{|A| - 1} \right)}}$$

It is a standardized measure of the deviation of the mean of C_q individuals from the general average. Using the central limit theorem we can add that

$$\bar{V}_q \rightarrow N \left(\bar{v}, \sqrt{\frac{\sigma^2}{|C_q|} \left(\frac{|A| - |C_q|}{|A| - 1} \right)} \right)$$

Thus, the v-test approximates towards a statistic Z-test of the null hypothesis *the average of X for cluster C_q is equal to the general average* with a p-value associated. If a variable rejects this hypothesis (with a p-value < 0.05), then its average for C_q differs from the general average, thus characterizing the cluster C_q . In the end, we keep for each cluster the variables that reject the null hypotheses to define each cluster.

A.6 Chilean trajectories

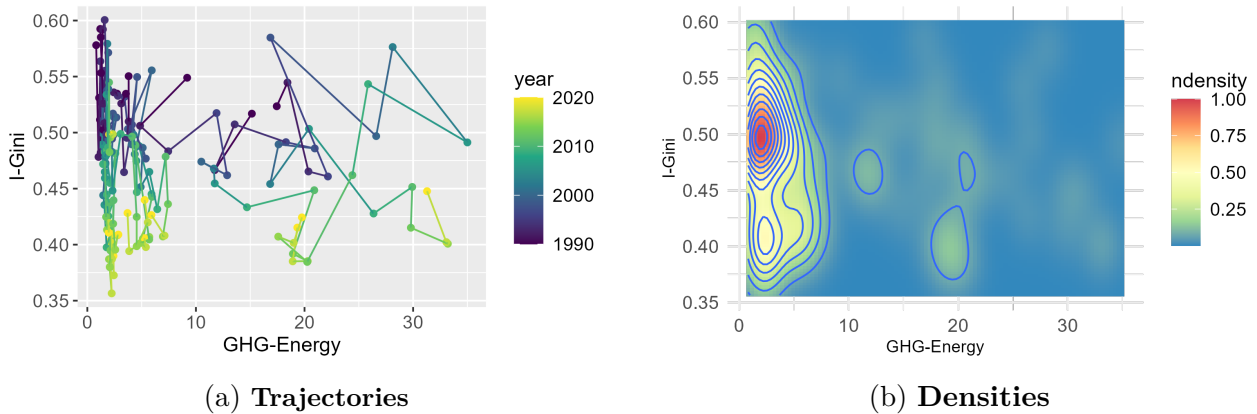


Figure 11: Trajectories (a) and densities (b) of the regions of Chili in the two-dimensional space of Gini in function of GHG emissions (in the energy sector). The energy sector is one of the main sectors of emissions for Chile.

A.7 Density map of France

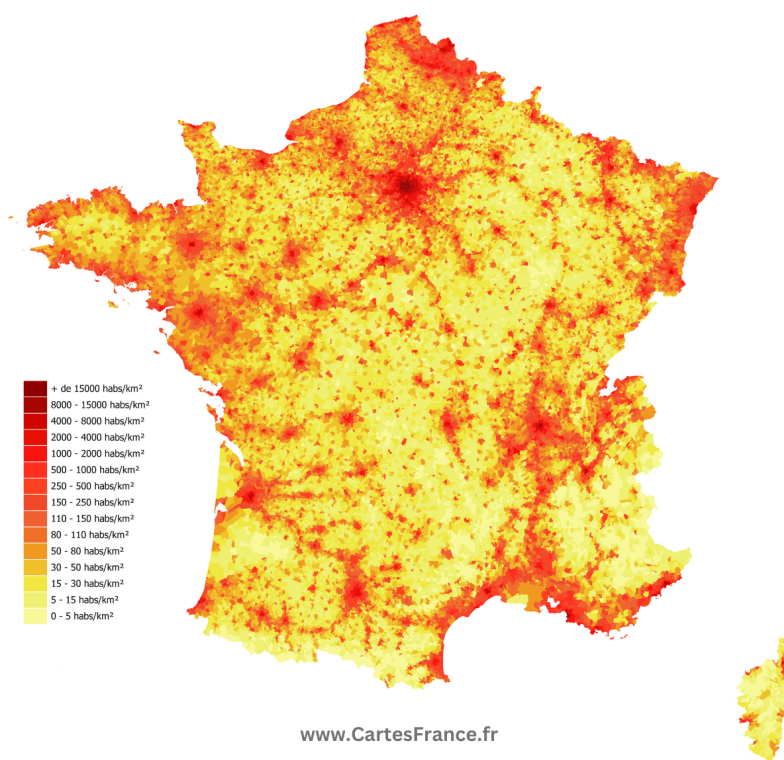


Figure 12: Maps of the population density of France (2009)